Viscosimetry

Protocol for the laboratory course in Polymer chemistry

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Stuttgart, April 14, 2025

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Date: 13 march 2025

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1 Introduction and Theory [1]

The viscosimetry is one method to determine the average molar mass of a polymer. It uses the fact, that the viscosity of a polymer solution is strongly dependent on the molar mass. Another important factor for the viscosity is the chemical nature of the polymer and how it interacts with the solvent. Therefore, the viscosimetry is not an absolute, but a relative method, which requires the measurement of standard solutions with defined molar masses dissolved in a specific solvent and at a specific temperature.

One kind of illustrating viscosity is the laminar movement of a solution in a capillary. Due to friction the laminar velocity is smaller, the nearer the liquid is to the vessel wall and gets maximized in the middle between the two vessel wall. This follows an overall speed of the liquid, which is antiproportional to the viscosity. The polymer balls are increasing the viscosity, because they are way bigger than the solvent molecules and connect the laminar layers of different velocities in the capillary with their chains. This leads to a decrease in the velocity, because the layers nearer to the walls are slowing the faster layers down. For quantitative determination of the viscosity the Hagen-Poiseuille equation (equation 1.1) is used.

$$\eta = \frac{\pi \cdot r^4 \cdot t \cdot \Delta p}{8 \cdot l \cdot V} \tag{1.1}$$

The absolute viscosity η is calculable with the radius of the capillary r, the length of the capillary l and its volume V. More important is the time t a liquid needs to cover a specific distance in the capillary. $\Delta p = \rho \cdot g \cdot h$ is the pressure difference at the capillary it is dependent of the height of the capillary h and the density of the solution ρ. In the practical approach a time is measured in which a solution covers a specific distance in the capillary.
For polymer analytical purpose the absolute viscosity of the solution is of no interest, rather than the relative viscosity η_{rel} compared to the solvent.

$$\eta_{\text{rel}} = \frac{\eta}{\eta_0} \tag{1.2}$$

In there η is the viscosity of the polymer solution and η_0 is the viscosity of pure solvent. With the relative viscosity it is possible to calculate the specific viscosity η_{sp} as equation 1.3 shows.

$$\eta_{\text{sp}} = \eta_{\text{rel}} - 1 \tag{1.3}$$

If the Hagen-Poiseuille equation is applied to equation 1.2 and 1.3 an expression for the specific viscosity, which is only dependent on the flow time t and the density of the solution ρ is received in equation 1.4. This is because in both the pure solvent and the polymer solution the same constants of the capillary are used, so they cancel out.

$$\eta_{\text{sp}} = \frac{t \cdot \rho - t_0 \cdot \rho_0}{t_0 \cdot \rho_0} \tag{1.4}$$

At low concentrations of the polymer it is possible to simplify this equation by assuming, that the density of the polymer solution is nearly the same as the density of the solvent.

This leads to an expression of the specific viscosity shown in equation 1.5.

$$\eta_{\mathrm{sp}} \approx \frac{t - t_0}{t_0} \tag{1.5}$$

Another important dimension is the Staudinger index $[\eta]$ it is defined in the following equation. With η_{red} being the reduced viscosity.

$$[\eta] = \lim_{c \to \infty} (\eta_{\mathrm{red}}) = \lim_{c \to \infty} \left(\frac{\eta_{\mathrm{sp}}}{c} \right) \tag{1.6}$$

So it is possible to gain the Staudinger index of a polymer-solvent system at a specific temperature, by conducting a concentration series, plotting the data of the reduced viscosity against the concentration and extrapolating for an infinitely low concentration. In the Mark-Houwink equation (equation 1.7) the connection between the Staudinger index and the viscosity average of molar mass $\bar{\mathrm{M}}_\eta$ is shown.

$$[\eta] = K \cdot \bar{\mathrm{M}}_\eta^a \tag{1.7}$$

K and a are the Mark-Houwink coefficients. They are only valid for one polymer-solvent system at one temperature. They are obtained by measuring the flow time of a concentration series with a standard polymer solution with a known molar mass. By a graphic display of the linearized Mark-Houwink equation, both a and K are easily determinable. For many common polymers these Mark-Houwink coefficients are enlisted in literature for many solvents.
Furthermore, the coefficient a gives information about the conformation of the polymer in the solution. For $a = 0$ the polymer behaves like a compact ball in the solution. For $a = 2$ the polymer can be seen as a rigid rod. And for $a = 0.5$ the polymer is in the thermodynamically ideal θ-state.

2 Experimental conduction

With viscosimetry it is possible to determine the viscosity average of molar mass. In this work the average molar mass of a polystyrene sample was determined. This was done by measuring the flow time of the solvent and the polymer solutions of different concentrations. To ensure the same measurement conditions for each measurement, an Ubbelohde viscometer was used. A schematic illustration of an Ubbelohde viscometer can be seen in figure 1.

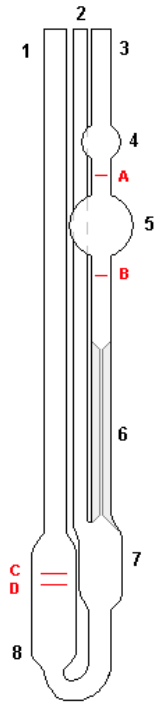

Figure 1: Schematic overview of an Ubbelohde viscometer [2]

The investigated liquid was filled into the reservoir (**8**). By applying a vacuum the liquid was then first brought to the level bulb (**7**) and through the capillary (**6**) the level of the solution was brought up to the feeder bulb (**4**). It was then vented. The level of the solution was sinking and the time it took the solution to sink from marking **A** to marking **B** was measured. This is the flow time. For ensuring that all measurements were done at the same pressure a pressure equalization tube (**2**) was used. Because temperature has a significant influence on the viscosity, all the measurements were done in a tempered water bath at 30 °C.

First the flow time of pure toluene was measured. This was done five times. Afterward the apparatus was cleaned with acetone for three times. Then this procedure was repeated with polystyrene solutions of four different concentrations. These concentrations were $c_1 = 1.382\,\mathrm{mg/mL}$, $c_2 = 1.980\,\mathrm{mg/mL}$, $c_3 = 5.394\,\mathrm{mg/mL}$ and $c_4 = 10.88\,\mathrm{mg/mL}$.

3 Analysis

The measured flow times of the pure toluene and the polymer solutions of different concentrations are enlisted in table 1.

Table 1: Measured flow times of solutions of different concentrations of polystyrene

$c\,[\mathrm{mg \cdot ml^{-1}}]$	t_1 [s]	t_2 [s]	t_3 [s]	t_4 [s]	t_5 [s]
0	23.39	23.39	23.39	23.42	23.39
1.382	25.10	25.10	25.10	25.10	25.10
1.980	25.88	25.88	25.88	25.89	25.89
5.394	30.76	30.77	30.78	30.74	30.77
10.88	40.93	40.93	40.95	40.95	40.96

Form these times the arithmetic mean of the flow time $\bar{t}$ was calculated for the used concentrations. With this time the specific viscosity is calculable by using equation 1.5. In the following this is done for the lowest concentration of polystyrene.

$$\eta_{\mathrm{sp}} = \frac{\bar{t_1} - \bar{t_0}}{\bar{t_0}} = \frac{25.10\,\mathrm{s} - 23.396\,\mathrm{s}}{13.396\,\mathrm{s}} = 0.0728$$

With this a value for the reduced viscosity is obtainable, like the following calculation shows.

$$\eta_{\mathrm{red}} = \frac{\eta_{\mathrm{sp}}}{c} = \frac{0.0728}{1.382\,\mathrm{mg \cdot ml^{-1}}} = 0.0527\,\mathrm{ml \cdot mg^{-1}}$$

The calculated values of $\bar{t}$, η_{sp} and η_{red} are enlisted in table 2.

Table 2: Mean flow time, specific and reduced viscosity for every measured polymer solution.

$c\,[\mathrm{mg \cdot ml^{-1}}]$	$\bar{t}$ [s]	η_{sp}	$\eta_{\mathrm{red}}\,[\mathrm{ml \cdot mg^{-1}}]$
0	23.396	0	-
1.382	25.100	0.0728	0.0527
1.980	25.880	0.106	0.0537
5.394	30.760	0.315	0.0584
10.88	40.930	0.750	0.0689

The graphic display of η_{red} in dependence of c is shown in the following figure 2.

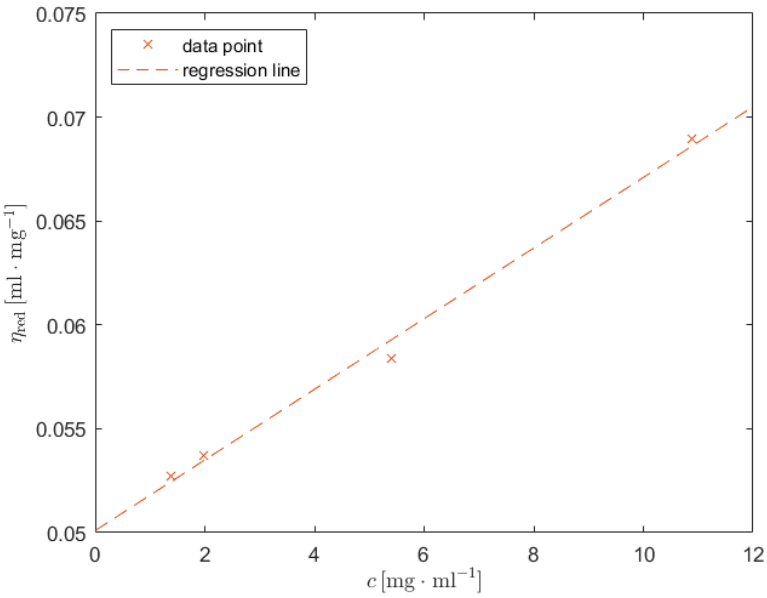

Figure 2: Dependence of η_{red} and concentration with a regression line.

For the regression line the following equation is valid.

$$\eta_{\text{sp}}/c = 0.0017\,\frac{\text{mL}^2}{\text{mg}^2}\cdot c + 0.0501\,\text{ml/mg} \tag{3.1}$$

The value of the reduced viscosity for an infinite low concentration is equal to the Staudinger index $[\eta]$. From equation 3.1 follows for $c = 0\,\text{mg/mL}$ that $[\eta] = 0.0501\,\text{ml/mg}$. From the Mark-Houwink equation (equation 1.7) with the Staudinger index the viscosity average of molar mass of the polymer can be calculated, if a and K are known. For polystyrene in toluene these values are known. They have the value of $a = 0.71$ and $K = 0.000\,012\,\text{ml/mg}$.

$$\bar{\text{M}}_\eta = \sqrt[a]{\frac{[\eta]}{K}} = \sqrt[0.71]{\frac{0.0501\,\text{ml/mg}}{0.000\,012\,\text{ml/mg}}} = 125\,752.37\,\frac{\text{g}}{\text{mol}}$$

So the average of the molar mass by viscosity of polystyrene is $\bar{\text{M}}_\eta = 125\,752.37\,\text{g/mol}$.

4 Discussion

In viscosimetry it is important to neither contaminate the investigated solution with other liquids, nor to contaminate it with solids. Therefore, the solution should be filtrated to separate any non dissolved solids. This was not done in this work and may be one source for deviations. Another might be leftover acetone in the apparatus. It was tried to remove all the acetone in the apparatus but for obvious reasons this was not possible to do completely. So this acetone might have influenced the viscosity of the investigated solutions.

5 Questions

1. Why and how do the Staudinger indices from polystyrene, polystyrenyl-lithium and sulfonated polystyrene differ?

In figure 3 the structure of polystyrenyl (1), polystyrenyl-lithium (2) and sulfonated polystyrene (3).

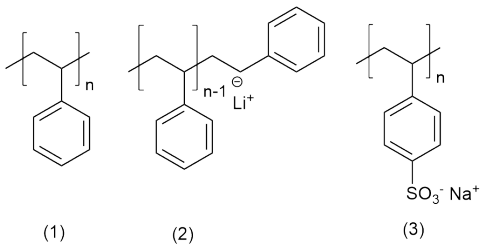

Figure 3: Structure of polystrene (1), polystyrenyl-lithium (2) and sulfonated polystyrenyl (3).

In polystyrenyl-lithium one end of the polymer chain is lithiated. Because of that ionic character of the end of the chain, the polymer can not entangle into a compact ball like in polystyrene. Therefore, the hydrodynamic radius is bigger, leading to a higher value for a and following from equation 1.7 to a higher Staudinger index. The same behavior is observed for sulfonated polystyrene. In this polymer every side chain has an ionic sulfone group. The ability of the polymer chains to entangle is even more hindered. The value for a is even higher than for polystyrenyl-lithium and so is the Staudinger index for sulfonated polystyrene drastically higher than for polystyrene and polystyrenyl-lithium.

2. Which factors can not be neglected for the determination of the Staudinger index, when the polymer concentration is above 2×10^{-2} g/mL?

With polymer concentrations of more than 2×10^{-2} g/mL the polymer chains start to become looped into each other. The polymer chains than, can no longer be seen as separate entangled balls. This leads to an even higher viscosity than expected, because the looped chains develop a complex network and interacting even more with the laminar layers in the solvent. Another point is, that with these concentrations the density of the polymer solution differ from the density of the solvent. Therefore, the simplification made to obtain 1.3 is no longer valid.

3. What is the difference between an Ostwald and an Ubbelohde viscometer?

In contrast to the Ostwald viscometer the Ubbelohde viscometer has a pressure equalization tube. Therefore, inaccuracies due to differences in the pressure can be minimized. This is important for liquids of high viscosity. Because of that the Ubbelohde viscometer is also suitable for solutions of higher viscosity, while the Ostwald viscometer is just for low viscose solutions. Other difference is the handleability. For correct measurements the Ostwald viscometer has to be filled with the same volume in each measurement. This is not the case with the Ubbelohde viscometer, which makes it more handleable.

4. Which solution has a higher Staudinger index? it-PMMA/heptanone-3 or at-PMMA/heptanone-3 at the same molar mass and identical measurement

conditions.

In figure 4 the structure of it-PMMA (1) and at-PMMA are depicted.

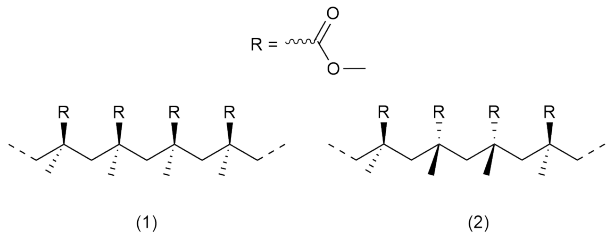

Figure 4: Structure of it-PMMA (1) and at-PMMA (2).

The important factor regarding the Staudinger index and the structure of a polymer is the Mark-Houwink coefficient a. Because it-PMMA features the same configuration of the side chain in every repetition unit, the chains are more stiff, than in at-PMMA. Therefore the coils of the chain in it-PMMA are less expanded than in at-PMMA. The larger coil leads to a higher hydrodynamic radius, which results in a higher value for a in at-PMMA. Following from the Mark-Houwink equation a higher value for a leads to a higher Staudinger index

6 References

[1] *Script to "laboratory course in polymer chemistry"*, University of Stuttgart, **2025**.

[2] *Visco Handbook:Basics and Application of Viscometry Using Glass Capillary Viscometers*, Xylem Analytics Germany Sales GmbH & Co. KG, **2025**.